



# PROGRAMMING WITH PYTHON

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## Python Basics and Standard Library Recursive Functions

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## Recursion

- Recursion is the process of defining something in terms of itself.
- In 1960 Dutch computer scientist E W Dijkstra wrote a paper on how to do recursion on a computer.



Fig: [https://en.wikipedia.org/wiki/Edsger\\_W.\\_Dijkstra](https://en.wikipedia.org/wiki/Edsger_W._Dijkstra)



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## Recursion

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- A function can call other functions.
- It is even possible for the function to call itself.
- These type of construct are termed as recursive functions.



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## Recursion

- In programming, two steps are required.
  - Express solution for a problem in terms of the problem itself, but of a smallest size
    - This is called the recursive call.
  - Express solution to a simple or base case without recursion
    - This is called base case or termination case.



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## Recursion

- Sum of  $n$  numbers can be written as  $n + \text{sum}(n-1)$ .

```
def snn(n):  
    if n == 0:  
        return 0  
    else:  
        return n + snn(n - 1)
```

```
n = 5  
print(f"The sum of the first {n} numbers is:", snn(n))
```

The sum of the first 5 numbers is: 15



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## Recursion

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- Recursion can cause stack overflow issues.
- Python has a built-in recursion limit.
- This limit prevents infinite recursion from causing an overflow of stack and crashing Python.



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## Recursion

- Factorial of  $n$  can be expressed as  $n$  times factorial of  $(n - 1)$  if  $n > 0$  and factorial of 0 can be expressed as 1.

```
def factorial(n):  
    if n == 0 or n == 1:  
        return 1  
    else:  
        return n * factorial(n - 1)
```

```
n = int(input("Enter the number"))  
print(f"The factorial of {n} is:", factorial(n))
```

```
Enter the number 7  
The factorial of 7 is: 5040
```

```
n = int(input("Enter the number"))  
print(f"The factorial of {n} is:", factorial(n))
```





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## Recursion

- Given two numbers the gcd is calculated as
  - If the second number becomes 0, return the first number as gcd.
  - Else recursively call the function with the arguments as the second number and the remainder when the first number is divided by the second number.



# PROGRAMMING WITH PYTHON

## Recursion

- Given two numbers the gcd is calculated as
  - If the second number becomes 0, return the first number as gcd.
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## Recursion

```
def gcd(a, b):  
    if b == 0:  
        return a  
    else:  
        return gcd(b, a % b)
```

```
num1 = int(input("Enter the first number"))  
num2 = int(input("Enter the second number"))  
print(f"The GCD of {num1} and {num2} is:", gcd(num1, num2))
```

Enter the first number 10

Enter the second number 2

The GCD of 10 and 2 is: 2



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## Recap

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- Recursion
- Examples



**THANK YOU**

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